
EXECUTIVE CREDIT RISK REPORT

Mortgage Portfolio Credit Risk Analysis

Fannie Mae Single-Family Loan Performance Dataset | Reporting Period: 2020 Q1-Q2

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Classification	Executive — CRO and Credit Risk Committee Distribution
Data Source	Fannie Mae SF Loan Performance Dataset — Latest Portfolio Snapshot
Portfolio Scope	Approximately 1.92 Million Unique Loans 2020 Q1-Q2

This report is prepared for internal executive use. Findings and recommendations are based exclusively on verified portfolio analytics derived from the Fannie Mae Single-Family Loan Performance Dataset.

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1 EXECUTIVE SUMMARY

This report presents the findings of a comprehensive mortgage portfolio credit risk assessment conducted using the Fannie Mae Single-Family Loan Performance Dataset for the first half of 2020. The analysis was performed on a latest-snapshot portfolio of approximately 1.92 million unique loans and examined five dimensions of portfolio risk: borrower credit quality, loan structure, property characteristics, geographic concentration, and behavioural distress.

The portfolio demonstrates generally sound performance. The delinquency rate stands at 0.65%, with 99.3% of loans performing as contractually expected. Total portfolio exposure amounts to \$194 billion across the snapshot population, with serious delinquency contained at 0.22%.

Despite this overall resilience, the analysis identified concentrated areas of elevated risk that warrant structured management attention. Modified loans represent less than 1% of total portfolio exposure but exhibit a delinquency rate of 22.8%, confirming modification status as the single strongest behavioural predictor of credit distress within the portfolio. Borrowers in the highest credit-risk segment record delinquency rates more than twenty times above those in the lowest-risk segment. Geographic concentration in California, Texas, Florida, and New York creates exposure to regional economic disruption, and New York exhibits elevated delinquency across multiple independent risk dimensions.

The portfolio surveillance framework established in this review identifies Modified Loans as the only monitoring segment currently in breach of defined risk thresholds. All other monitored segments — High Risk Borrowers, Purchase Loans, New York Delinquency Rate, and Serious Delinquency — remain within established limits.

Overall assessment: The portfolio is fundamentally healthy. Risk is identifiable, concentrated in specific segments, and manageable through targeted surveillance. The findings support a maintain-and-monitor posture with enhanced oversight of modification activity.

2 BUSINESS CONTEXT

The reporting period — January through June 2020 — coincides with the onset of the COVID-19 pandemic. During this period, the United States mortgage market experienced unprecedented policy interventions, including federal forbearance programmes, payment deferral arrangements, and foreclosure moratoria.

These interventions materially influenced observable delinquency and foreclosure behaviour during the period. Borrowers in active forbearance were classified as current under the terms of their assistance agreements, which suppresses reported delinquency rates below levels that would otherwise reflect underlying borrower financial stress. Foreclosure activity during 2020 Q1–Q2 was near-zero, reflecting the foreclosure moratorium rather than an absence of distressed loans progressing toward loss.

The portfolio metrics presented in this report therefore reflect contractual loan performance rather than the full extent of underlying borrower financial stress. Forbearance and deferral programmes active during this window represent a form of policy-assisted cure that may not be available in future stress periods. Re-delinquency risk among assisted borrowers cannot be assessed from the current snapshot alone and warrants continued monitoring as these programmes unwind.

3 PROJECT OBJECTIVES

This review was conducted to address the following executive business questions:

- Which borrower characteristics are most strongly associated with mortgage delinquency, and which segments require enhanced monitoring?
- Which loan and property structures carry elevated structural risk within the portfolio?
- Where is portfolio exposure geographically concentrated, and do high-exposure markets also exhibit elevated credit risk?
- How does loan modification status affect delinquency severity, and is this effect independent of existing borrower risk classification?
- Which portfolio segments require active surveillance, and which currently exceed defined risk thresholds?

The analysis was structured to support decision-making across four stakeholder groups: underwriting teams requiring risk driver evidence to inform approval criteria; credit risk teams requiring segment-level monitoring intelligence; portfolio managers requiring exposure concentration and diversification analysis; and executive leadership requiring a consolidated portfolio health assessment and actionable risk recommendations.

4 DATA AND METHODOLOGY

Data Source

The analysis is based on the Fannie Mae Single-Family Loan Performance Dataset for 2020 Q1 and Q2. This dataset covers Fannie Mae-acquired conforming conventional single-family mortgages across the United States and contains origination attributes, monthly performance records, and loan resolution data.

Analytical Dataset

The source dataset comprised 84.18 million monthly loan-performance observations following data quality validation and cleaning. Records with missing origination credit scores were removed, totalling 40,190 records — 0.048% of the full dataset — with negligible impact on portfolio representation.

For analytical purposes, the integrated source dataset was separated into logical origination and performance layers within PostgreSQL. This supported feature engineering, exploratory data analysis, and portfolio KPI development across the full servicing history.

Reporting Architecture

Executive reporting and dashboard development utilise a latest-snapshot fact table of approximately 1.92 million unique loans. This model retains one record per loan, representing the most recent available servicing observation. The approach ensures that executive KPIs reflect the current state of the portfolio rather than cumulative historical balances, prevents double-counting across reporting periods, and supports accurate interactive analysis within the Power BI reporting environment.

Important: Portfolio exposure figures in this report represent the current outstanding unpaid principal balance of unique loans as of their most recent servicing observation. These figures reflect the current portfolio snapshot and should not be interpreted as cumulative historical balances.

Risk Feature Engineering

The following analytical risk features were derived from origination and performance attributes to support segmentation and delinquency analysis:

Feature	Definition	Risk Basis
Credit Score Band	Very Low / Low / Moderate / High / Very High Risk	Strongest single PD predictor
LTV Band	Very Low (<60%) / Low (60–80%) / Moderate (80–90%) / High (>90%)	Collateral coverage and loss severity
DTI Band	Low (<28%) / Moderate (28–36%) / High (36–43%) / Very High (>43%)	Payment capacity indicator
Risk Factor Count	Count of risk conditions: Credit Score <680, LTV >90, DTI >43	Cumulative risk exposure measure
Risk Segment	Low Risk (0 factors) / Moderate (1 factor) / High (2–3 factors)	Multi-variable risk layering
Delinquency Flag	Delinquency Status ≠ 00	Identifies all delinquent observations
Serious Delinquency Flag	Delinquency Status ≥ 03 (90+ DPD)	Identifies loans approaching default

5 EXECUTIVE FINDINGS

5.1 Portfolio Health

The portfolio demonstrates sound overall credit quality across its 1.92 million unique loans and \$194 billion in current outstanding exposure. The following metrics characterise portfolio health at the time of this review:

Total Portfolio Exposure	\$194 billion	Current outstanding UPB — 1.92M unique loans
Portfolio Delinquency Rate	0.65%	30+ days past due as share of total loans
Serious Delinquency Rate	0.22%	90+ DPD — default proximity indicator
Current Performing Rate	99.3%	Loans with current payment status
Delinquent Exposure	\$2.6 billion	Outstanding UPB of delinquent loans
Total Unique Loans	1.92 million	Latest-snapshot portfolio count

The vast majority of the portfolio — 99.3% — remains contractually current. Serious delinquency at 0.22% is well below the 0.5% surveillance threshold established for this portfolio. Foreclosure activity during the reporting period was effectively zero, reflecting the federal foreclosure moratorium active during 2020 Q1–Q2 rather than an absence of underlying borrower stress.

The absolute scale of the portfolio amplifies the financial impact of even modest credit deterioration. A one percentage point increase in the delinquency rate would move approximately \$1.94 billion in additional exposure into delinquent status, reinforcing the importance of early identification and targeted segment-level intervention.

5.2 Borrower Risk Profile

Borrower credit quality is the strongest individual predictor of mortgage delinquency across the portfolio. Delinquency rates increase consistently and materially as credit quality deteriorates, with a more than twenty-two-fold difference in delinquency rates between the highest and lowest credit score segments.

Credit Score Band	Loan Count	Delinquency Rate	Assessment
Very Low Risk	1.13M	0.2%	Well below portfolio average — strong performance
Low Risk	0.45M	0.7%	Near portfolio average — standard monitoring
Moderate Risk	0.25M	1.5%	Above average — targeted oversight warranted
High Risk	0.07M	2.9%	Elevated — enhanced monitoring required
Very High Risk	0.01M	4.4%	Highest risk — priority surveillance segment

The portfolio's borrower composition provides a structural buffer: 1.13 million loans — nearly 59% of the portfolio — reside in the Very Low Risk credit segment. However, the risk gradient is steep. Very High Risk borrowers are more than twenty-two times more likely to be delinquent than Very Low Risk borrowers.

Debt-to-income ratio demonstrates a consistent but moderate relationship with delinquency, with rates rising from 0.3% in the Low Risk DTI band to 1.0% in the Very High Risk band. DTI functions as a secondary risk indicator, most valuable when combined with credit score in the risk-layering framework.

Risk layering analysis reveals that combining multiple risk conditions amplifies delinquency risk non-linearly. Loans exhibiting two or more concurrent risk conditions record delinquency rates of 2.4%, nearly three times the 0.3% rate among low-risk loans. Borrower risk assessment must evaluate the interaction of risk factors collectively rather than individual metrics in isolation.

The Moderate Risk segment warrants particular attention. Despite lower individual delinquency rates than the High Risk segment, Moderate Risk loans account for the largest share of delinquent exposure at \$1.3 billion, reflecting their significantly larger portfolio volume. This underscores that effective exposure management requires weighting both delinquency rates and outstanding balances rather than risk rate alone.

5.3 Property and Loan Characteristics

Loan purpose is the primary structural risk driver within the portfolio. Purchase loans record a delinquency rate of 1.1%, compared with 0.6% for Cash-Out Refinance loans — approximately double the rate. This differential is consistent with two factors specific to the 2020 Q1-Q2 period: purchase loans originated in early 2020 were exposed to the COVID-19 income shock during the earliest months of their lifecycle, when borrower equity is lowest and historical delinquency risk is highest. Vintage-adjusted analysis across subsequent quarters will be required to determine whether this reflects a temporary timing effect or a persistent structural risk.

Loan-to-value ratio demonstrates a consistent positive relationship with delinquency risk, with rates rising from 0.4% in the Very Low Risk LTV band to 1.4% in the High Risk band. Portfolio concentration in the lower LTV bands — 0.42 million loans in the Very Low Risk band and 0.74 million in the Low Risk band — provides a collateral quality buffer against widespread loss severity in a stress scenario.

Manufactured Housing exhibits the highest property-type delinquency rate at 1.8%, compared with 0.7% for Single Family and 0.5% for Condominiums. Standard Single Family homes, the dominant property type, record delinquency in line with the portfolio average. The Manufactured Housing finding warrants additional underwriting scrutiny, though its limited portfolio share constrains overall financial impact.

5.4 Geographic Risk Analysis

Portfolio exposure is geographically concentrated. California, Texas, Florida, Washington, and Colorado collectively represent the largest share of outstanding balances, with California alone accounting for approximately \$32.3 billion — the single most significant geographic concentration within the portfolio.

State	Exposure	Loans	High Risk %	DQ Rate	Serious DQ	Modified %
California	\$32bn	301K	3.20%	0.47%	0.15%	0.43%
Texas	\$14bn	134K	6.76%	1.05%	0.37%	0.68%
Florida	\$11bn	110K	7.14%	0.89%	0.35%	0.71%
New York	\$7bn	42K	4.72%	1.18%	0.41%	1.03%
Washington	\$8bn	72K	3.63%	0.41%	0.13%	0.31%
Colorado	\$8bn	76K	3.62%	0.39%	0.14%	0.29%
Virginia	\$6bn	50K	4.34%	0.51%	0.18%	0.37%
Illinois	\$6bn	74K	4.39%	0.73%	0.27%	0.42%
New Jersey	\$6bn	46K	4.47%	0.71%	0.21%	0.58%

State	Exposure	Loans	High Risk %	DQ Rate	Serious DQ	Modified %
Georgia	\$6bn	60K	5.14%	0.80%	0.24%	0.53%

A key diagnostic finding of the geographic analysis is that portfolio risk is driven primarily by borrower composition rather than location alone. States exhibiting elevated delinquency consistently show higher concentrations of High Risk borrowers and modified loans, confirming that geographic risk reflects differences in underlying credit quality rather than regional market dynamics in isolation.

New York presents the most concerning geographic risk profile, recording the highest delinquency rate (1.18%) and the highest modification concentration (1.03%) among all major markets. These elevated readings persist across multiple independent risk dimensions — credit quality, delinquency severity, and behavioural distress — suggesting systemic borrower-quality challenges in this market.

California, despite representing the largest single-state exposure, demonstrates relatively strong underlying credit quality: a High Risk borrower share of 3.20% — the lowest in the diagnostic matrix — and a delinquency rate of 0.47%, well below the portfolio average. California's concentration reflects market scale rather than credit deterioration, though the concentration itself warrants scenario-based monitoring.

Florida and Texas combine meaningful exposure with above-average delinquency rates and elevated High Risk borrower concentrations (7.14% and 6.76% respectively), positioning both states as priority markets for geographic risk monitoring.

5.5 Behavioural Risk Analysis

Loan modification status is the single strongest behavioural predictor of credit distress identified across this analysis. Modified loans exhibit a delinquency rate of 22.8%, compared with 0.6% for non-modified loans — an approximately 38-fold differential that represents the most pronounced risk differentiation observed across any dimension of the portfolio.

Methodology note: The 22.8% delinquency rate for modified loans is derived from the latest-snapshot portfolio model, which captures each loan's current performance status at its most recent servicing observation. This reflects the current delinquency profile of the modified loan population within the portfolio at the time of analysis.

Modified loans represent less than 1% of total portfolio exposure — with outstanding modified loan balances of approximately \$0.3 billion — yet account for \$1.6 billion in delinquent exposure. This disproportionate contribution confirms that modification status identifies a highly concentrated pocket of behavioural distress within an otherwise well-performing portfolio.

The elevated delinquency associated with modification status persists across all borrower risk segments. Among High Risk borrowers, modified loans exhibit a delinquency rate of 27.0% versus 2.0% for non-modified counterparts. Among Low Risk borrowers, modified loans record 17.8% delinquency versus 0.3% for non-modified counterparts. This cross-segment persistence confirms that modification status carries independent diagnostic value beyond existing borrower risk classification.

Loan modification is a response to pre-existing borrower distress rather than a cause of delinquency. The modification flag identifies loans that have already experienced financial difficulty sufficient to require formal loan restructuring. The elevated delinquency rate reflects underlying financial vulnerability and the associated risk of re-delinquency following modification.

5.6 Portfolio Surveillance

The portfolio surveillance framework consolidates the analytical findings from Sections 5.1 through 5.5 into an operational monitoring structure with defined risk thresholds and escalation status across five priority segments.

Monitoring Segment	Current Value	Risk Threshold	Variance	Status
Modified Loans	22.8%	20.0%	+2.8%	BREACH
High Risk Borrowers	2.4%	5.0%	-2.6%	NORMAL
New York	1.2%	1.5%	-0.3%	NORMAL
Purchase Loans	1.1%	1.5%	-0.4%	NORMAL
Serious Delinquency	0.2%	0.5%	-0.3%	NORMAL

Modified Loans represent the only segment currently exceeding its defined surveillance threshold, with a delinquency rate of 22.8% against a 20.0% risk limit. All other monitored segments remain within established boundaries, supporting an overall stable portfolio risk profile at the time of this review.

The monitoring priority framework ranks segments by composite risk severity: Modified Loans (Priority 1, index score 100), High Risk Borrowers (Priority 2, score 80), Purchase Loans (Priority 3, score 60), New York (Priority 4, score 55), and Serious Delinquency (Priority 5, score 40). Priority scores reflect relative surveillance urgency and do not imply that lower-ranked segments are without risk.

6 OVERALL PORTFOLIO RISK ASSESSMENT

The mortgage portfolio is fundamentally sound. With 99.3% of loans performing as contracted, low delinquency rates, and structural concentration in lower-risk borrower and collateral segments, the portfolio is well-positioned at the time of this review. The concentration of nearly 59% of loans within the Very Low Risk credit score band provides a meaningful buffer against broad-based credit deterioration.

Against this positive baseline, the analysis identifies three concentrated risk themes that characterise the portfolio's residual risk exposure:

Behavioural Distress Concentration

Modified loans, while representing a small share of total exposure, exhibit delinquency intensity approximately 38 times higher than the non-modified portfolio. This concentration is independent of borrower risk classification, making modification status a primary standalone risk indicator that the existing risk segmentation framework does not fully capture.

Geographic Risk Interaction

New York consistently exhibits elevated risk across delinquency rate, serious delinquency rate, and modification concentration simultaneously — a multi-dimensional risk profile not observed in other major markets. Florida and Texas also combine meaningful exposure with above-average delinquency, creating geographic clusters where regional economic deterioration would have a disproportionate portfolio impact.

Purchase Loan Performance Differential

Purchase loans record approximately double the delinquency rate of refinance loans. While consistent with vintage and seasoning effects specific to the 2020 COVID-period originations, the magnitude warrants monitoring through subsequent reporting periods to distinguish temporary timing effects from persistent structural risk.

Risk conclusion: The portfolio warrants a maintain-and-monitor posture with active escalation for the Modified Loan segment, which is the only priority segment currently in breach of defined surveillance thresholds. The overall risk profile is stable but sensitive to deterioration in the Modified Loan, geographic concentration, and purchase loan segments.

7 STRATEGIC RECOMMENDATIONS

The following recommendations are drawn directly from the risk findings presented in Section 5. Each is structured to address a specific concentration of risk identified during the review, with the supporting evidence and proposed management action presented together.

Recommendation 1 — Establish a Dedicated Modified Loan Monitoring Programme

Finding: Modified loans exhibit a delinquency rate of 22.8%, exceeding the 20.0% surveillance threshold and representing the only current breach in the portfolio monitoring framework. This risk persists independently across all borrower risk segments.

Action: Implement a dedicated monthly review programme for modified loans. Modified loans progressing beyond early-stage delinquency post-modification should be escalated to workout teams for case-level assessment. Re-delinquency rates should be reported separately by modification type and reviewed quarterly by the Credit Risk Committee.

Recommendation 2 — Strengthen Risk-Layered Borrower Oversight

Finding: Borrowers exhibiting multiple concurrent risk conditions record delinquency rates of 2.4%, nearly three times those of low-risk borrowers. The Moderate Risk segment accounts for the largest share of delinquent exposure due to its larger portfolio volume.

Action: Implement enhanced portfolio surveillance for Moderate and High Risk segments, incorporating both delinquency rate and exposure-weighted monitoring. Review compensating factor requirements for new originations with two or more concurrent risk conditions.

Recommendation 3 — Implement Vintage-Adjusted Purchase Loan Monitoring

Finding: Purchase loans record a delinquency rate of 1.1%, approximately double that of refinance loans. This differential may reflect COVID-timing effects specific to 2020 originations but requires monitoring to distinguish structural from temporary risk.

Action: Track purchase loan performance separately from refinance loans, disaggregated by origination vintage. If vintage-adjusted delinquency for purchase loans remains materially elevated beyond 2020 cohorts, initiate a review of purchase loan underwriting standards.

Recommendation 4 — Apply Geographic Stress Monitoring for Priority Markets

Finding: New York exhibits the highest delinquency rate (1.18%), highest modification concentration (1.03%), and elevated High Risk borrower concentration simultaneously. California represents 16.6% of total portfolio exposure. Florida and Texas combine above-average delinquency with meaningful exposure.

Action: Conduct periodic geographic stress reviews for California, New York, Florida, and Texas. For California, develop scenario analysis modelling the impact of a 10% and 20% house price decline on portfolio LTV distribution and expected delinquency uplift. For New York, increase monitoring frequency given its multi-dimensional risk profile.

Recommendation 5 — Continue Serious Delinquency Migration Surveillance

Finding: Serious delinquency stands at 0.22%, below the 0.5% surveillance threshold. The delinquency severity distribution shows limited progression into advanced stages from the current portfolio position.

Action: Maintain monthly monitoring of transitions into 90+ DPD status as a leading indicator of default risk, with particular attention to the Modified Loan and High Risk Borrower segments where delinquency progression carries the highest marginal risk impact.

8 PROJECT LIMITATIONS

The following limitations apply to the findings and recommendations presented in this report:

Limitation	Description	Impact on Analysis
COVID-19 Policy Effects	Forbearance programmes and foreclosure moratoria active during 2020 Q1-Q2 suppress observable delinquency and prevent foreclosure activity from materialising.	Delinquency rates understate true borrower financial stress. Foreclosure near-zero reflects policy context.
Snapshot Reporting Model	Executive KPIs are based on the latest servicing observation per loan. Historical delinquency transitions are not reflected in dashboard metrics.	Roll rate and cure rate analysis require longitudinal data not available from the snapshot model.
LGD Estimation	Insufficient foreclosure exit events during the moratorium period to support statistically reliable LGD estimation from actual recovery data.	Loss severity estimates are directional only and rely on industry benchmark assumptions.
Current LTV	Only origination LTV is available. Post-origination changes in property values are not captured.	LTV risk analysis reflects origination leverage. Current leverage may differ for older vintages.
Income and Employment Data	DTI is the only income proxy available. No direct income, employment status, or income stability data is captured.	Payment capacity analysis is limited to the DTI measure available at origination.
Geographic Scope	Analysis covers Fannie Mae-acquired conforming conventional loans only. Non-conforming, FHA, VA, and non-agency loans are excluded.	Findings reflect the conforming conventional segment and may not represent the broader US mortgage market.

9 CONCLUSION

The Fannie Mae mortgage portfolio assessed in this review demonstrates sound overall credit quality for the 2020 Q1–Q2 reporting period. With 99.3% of loans performing as contractually expected, a delinquency rate of 0.65%, and serious delinquency contained at 0.22%, the portfolio's headline performance reflects disciplined origination standards and a borrower base weighted toward lower-risk credit profiles.

The analysis reveals that portfolio risk is concentrated and identifiable rather than broadly distributed. Modified loans, high-risk layered borrowers, purchase loan originations, and selected geographic markets — particularly New York — account for the majority of elevated risk observations within an otherwise stable portfolio. This concentration is analytically advantageous: it enables targeted monitoring and intervention at the segment level, avoiding the need for broad portfolio-wide risk responses.

Modified loan performance represents the most material risk finding of this review. The 22.8% delinquency rate in this segment, combined with the independent persistence of this risk across all borrower classifications, positions modification status as a primary portfolio risk indicator that merits dedicated surveillance infrastructure. It is the only monitoring segment currently in breach of defined risk thresholds.

The portfolio surveillance framework established through this analysis provides a structured operational foundation for ongoing risk management. With defined thresholds, clear escalation logic, and a priority ranking across five key segments, the institution is positioned to detect emerging credit deterioration before it materialises into financial loss.

The period under review carries an inherent analytical constraint. The COVID-19 forbearance environment suppressed observable delinquency and prevented foreclosure from registering in the data during 2020 Q1–Q2. The metrics presented in this report should therefore be understood as a floor rather than a ceiling of underlying portfolio stress. Continued performance monitoring through subsequent quarters — as forbearance programmes unwind and borrower assistance measures normalise — will be essential to fully validating the portfolio's credit trajectory.

The findings of this review support maintaining the current portfolio posture while implementing the five targeted recommendations outlined in Section 7. The portfolio does not currently exhibit systemic deterioration; risk remains concentrated within identifiable segments rather than distributed across the broader book. Continued management focus on the Modified Loan segment, priority geographic markets, and purchase loan vintage performance will represent the highest-value risk intervention over the near term.

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